pfi

Python for Fixed Income

COURSE 8 · IMPLEMENTING AGENTIC AI

Case Study: A Risk Agent on the Synthetic Book

WEEK 1 · DAY 5

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: a risk agent on the synthetic book

Week 1, one card

Day 1	one request, one tool result
Day 2	the loop
Day 3	your bond functions as tools
Day 4	figures that cite their request
Day 5	all of it on a whole book

Today

1. holdings_tools: four tools over the book
2. The ground truth first, by pandas
3. The case study: eight questions
4. Every figure, twice
5. The workbook, the run log, git tag
6. What the agent shows, where it fails

Every model response in the notebook is a saved response, replayed. A live run can differ.

The rules for agents

What an agent produces: text, every number from a tool, checked twice today

Allowed names	Never called
the agent returned	the agent's opinion or view
the saved run	what the agent believes
the brief states	the agent's advice

A sort of the synthetic book by a computed number describes the file

Allowed names	Never called
DV01 per 100 of face	cheap, rich, or attractive
position DV01 in dollars	a trade idea or a recommendation
the largest DTS contributions in the file	overweight or underweight

How tools over a book work

The problem	A tool cannot hand the model the file: it is large and the model would have to read it
The closure	holdings_tools(frame) builds four inner functions that remember the frame
The result	Pandas arithmetic, returned as a small dict with units in its keys and the date inside
You control	The frame, the measures, and the weights: market value
What comes out	A dictionary of four tools, offered beside day 3's seven

holdings_tools: four tools over the book

Four tools, the frame inside

```
bond_lookup(cusip)

sector_totals(measure)

largest(measure, n)

portfolio_totals()
```

Two measures, market-value weights

position DV01 ($) = duration x MV / 10,000

= DV01 per 100 x par held / 100

DTS contribution = spread duration x OAS

x the bond's share of the book's MV

measure: an enum of four names

```
portfolio_totals: 200 bonds, market value $490,819,215.00, duration 6.254777 years, DTS 1064...
sector_totals, position DV01 in dollars, first three: {'Financials': 42434.459291, 'Transpor...
```

market value is clean price plus accrued: yes; the file's dts is spread duration x OAS: yes

Every setting of the four tools

Tool	Setting	Type	Default	What the model chooses
`bond_lookup`	`cusip`	string	required	the bond's CUSIP
`sector_totals`	`measure`	string	required	one of four measures: an enum
`largest`	`measure`	string	required	one of four measures: an enum
`largest`	`n`	integer	required	how many bonds to return
`portfolio_totals`	`(none)`	-	-	no argument: the book's totals

`measure: market_value_usd, par_held_usd, position_dv01_usd, dts_contribution`

Spread duration is not a measure: that is why no tool sorts by it. Tune first: the choices of measure.

How an answer key works

The problem	A check written after the run can be bent to agree with it
The key	Computed by pandas on the file, with the tools' weights, before any request
The comparison	Each figure against the key, at the figure's own decimals
You control	Which questions get a key, and the convention behind it
What comes out	Numbers for each question and, for some, CUSIPs

The ground truth first

```
financials 42,434.459291
dts10 36.949407, 34.913216, 26.541171, ... CUSIPs ['99066...
longest 16.042000 CUSIPs ['99053BAB9']
duration 6.254777
bond_dts 1,158.275000
healthcare 38,908,187.500000
top5 5,435,500.000000, 5,422,700.000000, 5,287,462.500000...
dv01_oas 59.000000 CUSIPs ['99053BAB9']
```

7 of 8

keys the tools return exactly, at 6 decimals

1 of 8

no tool sorts by spread duration

pandas computes every answer before any request, with the tools' market-value weights.

A key computed after the run can be bent to agree. A key decided first cannot.

The case study

Context

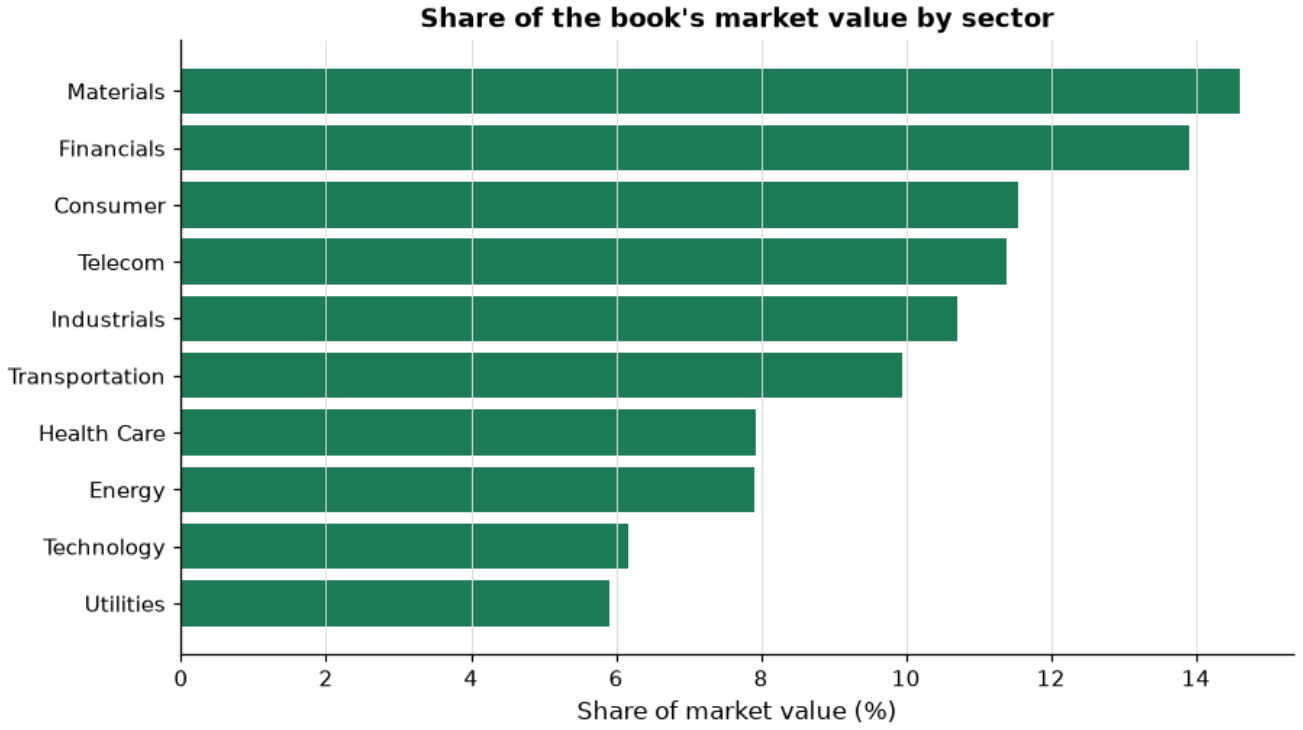
A risk desk asks the same questions every day: a sector's position DV01, the largest DTS contributions, the longest spread duration, the book's weighted duration. Can an agent answer them from the book's tools, and how do you know its numbers are right?

Objective

1. Eleven tools, eight descriptive questions
2. Every figure: against the run, and pandas
3. A workbook, runs.csv, git tag week1
4. What the agent shows, where it fails

holdings.csv, Course 1: 200 synthetic bonds, ten sectors as the file writes them ("Health Care").

The book by sector



Ten sectors, written as the file writes them: Health Care is two words. The bars describe the file; nothing here is a view on a sector.

How the risk agent works

The problem	Desk questions about a whole book, each answer checkable
The loop	Day 2's loop, day 4's answer format, eleven tools: seven bond and four holdings
The run	Descriptions read, tools requested one at a time, a DeskAnswer citing the requests
You control	The tools offered, the question, and the step limit of 8
What comes out	A run, a trace table to read first, an answer to check twice

Eight questions: what came back

Question	Tools requested	Figures	Traced	Agree with pandas
financials	sector_totals	1	1	1
dts10	largest	10	10	10
longest	largest then bond_lookup	1	1	1
duration	portfolio_totals	1	1	1
bond_dts	bond_lookup	3	3	0
healthcare	sector_totals	1	1	1
top5	largest	5	5	5
dv01_oas	largest then bond_lookup	1	1	1

23 figures: all 23 pass check_figures, 20 agree with the pandas key. The gap is the lesson.

How the two checks differ

The problem	A traced figure can be the wrong figure
check_figures	Follows the citation: the value of the result it cites, in its unit
agrees	Is the figure one of the numbers of the answer key?
You control	The key and the rounding
What comes out	One row per figure and one per question, kept in the workbook

Every figure, twice

check_figures: is the figure the value of the result it cites, at its rounding, in its unit?

agrees: is the figure one of the numbers of the answer key decided first?

23

figures in the eight answers

23

pass check_figures

20

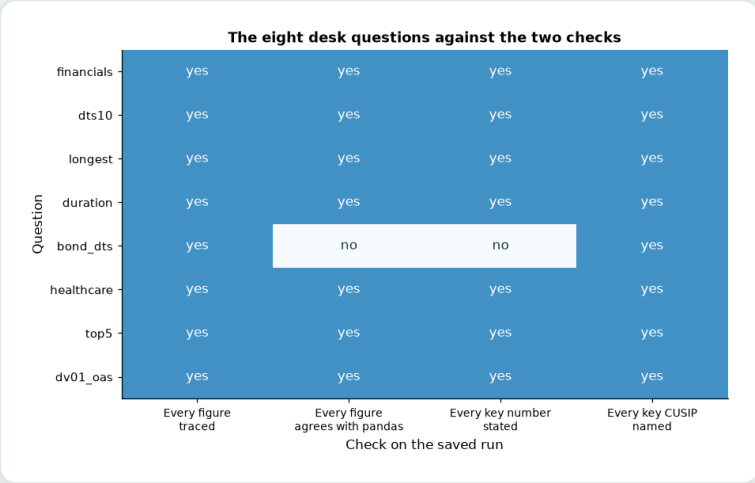
agree with the pandas key

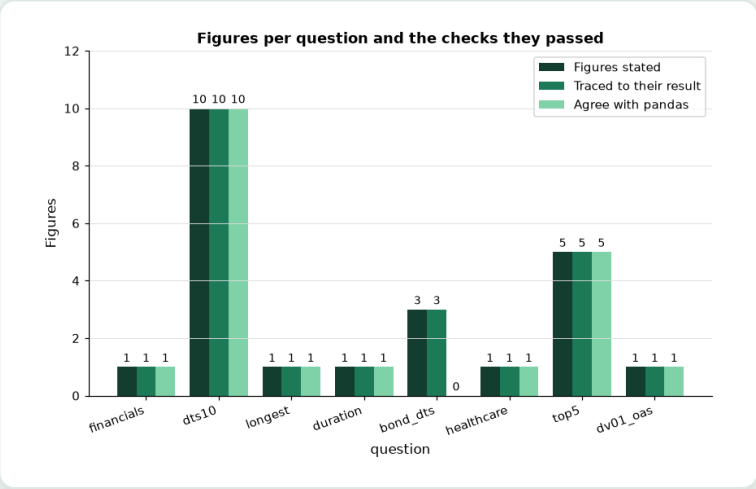
3

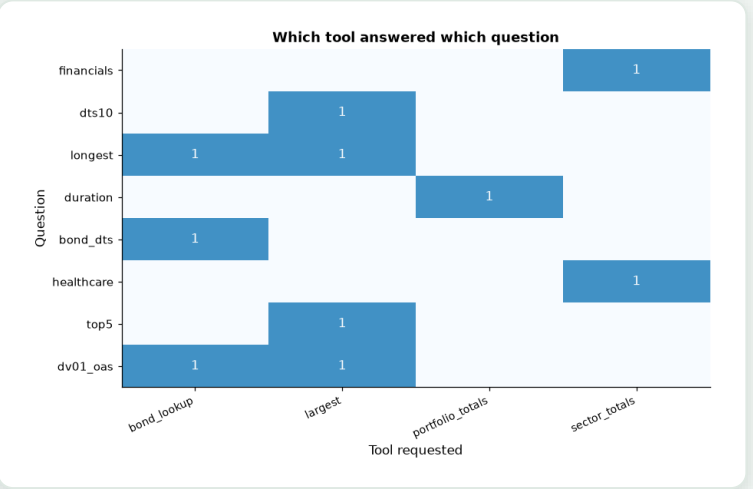
disagree with pandas, all from one answer

Every figure traces to its cited result, in its unit. The three that disagree with pandas trace too.

The eight questions against the checks





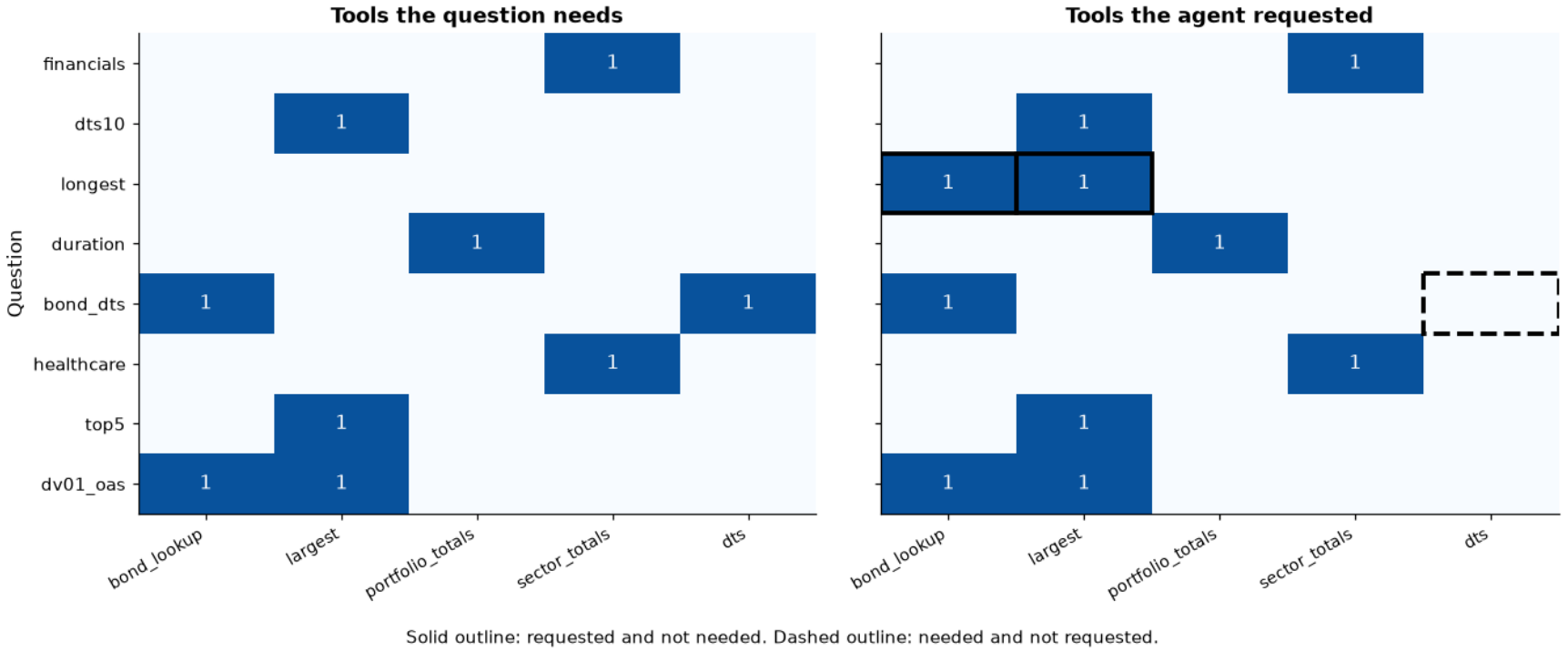


23 figures in the eight answers: 23 pass check_figures, 20 agree with the pandas key.

Read down a column for the check, along a row for the question, and across the tool grid for the tools it needed.

A light cell is a figure or a key the agent did not meet. The table says which; the grid shows where to look.

Tools needed against tools requested



Six of the eight match. The outlined cells are the two failures of section 5.8, visible before any figure is compared.

What the agent shows, and where it fails

What the agent did, step by step

```
1  assistant bond_lookup  {"cusip":"99049XAE2"} tool_calls 1637 20

1  tool bond_lookup  {"as_of": "2026-09-30", "clean_price_per_100": 97.556, "coupon_pct"...
```

1. Two tools needed, one requested: the bond's DTS came back as its dts_contribution

   ...and every check on the run passed it; only the pandas key fails it

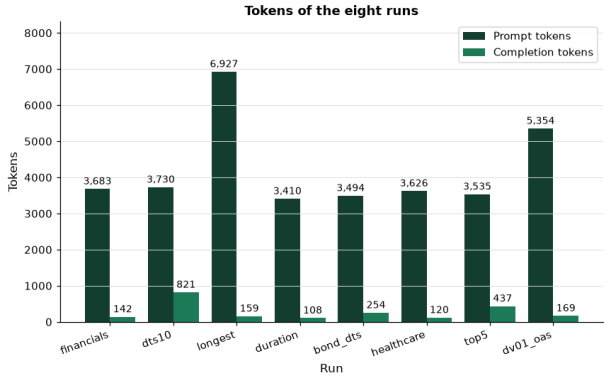
2. A right answer, no comparison made: longest spread duration found by sorting on DV01

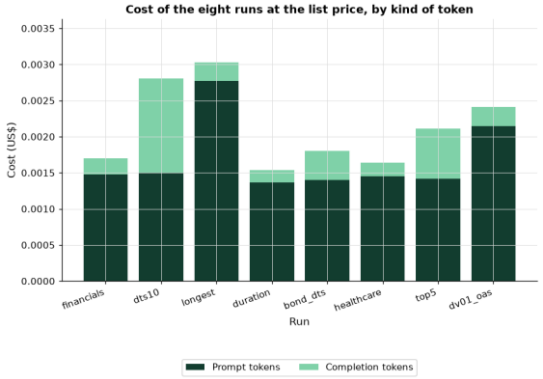
3. A name typed differently: "healthcare" mapped to Health Care, this time

4. The run log marks all 8 runs passed: a log of traces is not a grade

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 2 requests, 3,494 prompt and 254 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

What each run used and cost





Run	Requests	Prompt	Completion	Cost (US$)	Traced	Passed
financials	2	3,683	142	0.001700	1	True
dts10	2	3,730	821	0.002806	10	True
longest	3	6,927	159	0.003025	1	True
duration	2	3,410	108	0.001537	1	True
bond_dts	2	3,494	254	0.001804	3	True
healthcare	2	3,626	120	0.001642	1	True
top5	2	3,535	437	0.002113	5	True
dv01_oas	3	5,354	169	0.002412	1	True

Excel and Git

Excel side by side (tested in Excel)

`=SUMIFS(T2:T201,E2:E201,"Financials")`	42,434.459291
`=LARGE(U2:U201,1)`	36.949407
`=SUMPRODUCT(N2:N201,S2:S201)/SUM(S2:S201)`	6.254777
`=INDEX(O2:O201,MATCH("99049XAE2",B2:B201,0))*INDEX(M2:M201,MATC...`	1,158.275
`=INDEX(B2:B201,MATCH(MAX(O2:O201),O2:O201,0))`	99053BAB9
`=SUMIFS(S2:S201,E2:E201,"healthcare")`	0
`=SUMPRODUCT(N2:N201,R2:R201)/SUM(R2:R201)`	6.308362

Excel against pandas: 23 of 23 rows the same, within 1e-6

```
f475e2c (HEAD -> main, tag: week1) agentkit day 5: holdings_tools, 31 tests; runs.csv o...
```

runs.csv with every column filled, then git tag week1. 31 passed.

The AI check

The docstring is the prompt. The agent code that comes back runs and looks right.

1

Docstring

Market-value weights, as holdings_tools uses.

→

2

Code back

Weights by par_held. Runs, looks plausible.

→

3

Test first

Two bonds, near 62 and 119: must equal the tool. It fails.

→

4

Fix

Weight by market_value.

Today: the weights are par held, not market value: the tool's correct duration would look wrong.

Code that runs is not the same as code that is right.

What you can do now

- Tools over a whole book, weights in the docstrings
- Every answer key computed by pandas first
- Every figure checked twice: run and key
- Say where an agent's answer fails
- The same totals in Excel; git tag week1

Week 2, Day 6 Market data tools: the Treasury curve and FRED